
Library Management System

Introduction

Libraries offer a quiet space for study, to book lovers. Once a reader acknowledges that the book he had to resume studying from, has been taken by someone else, the quiet study space runs into a chaos.

Libraries have various management system specific to their needs; however, a majority of them work in offline mode; or at best, the interface is not accessible by the subscribers unless they manually verify their data each time they physically go to the library management window.

An online interface will help the subscribers track their usage and edit it real time, reducing the amount of work of the library management staff.

Purpose

If you need to subscribe to a library, you have to manually go to the library (this is how it works for majority of the cases) and get yourself registered as their user, pay the fee and grab a card for identification and tracking.

There are usually no online interfaces offered where the user/administration can track the activity.

In addition to this, a book; marked to be read at a later time; can possibly be at the hands of some other subscriber when the previous reader is searching for it.

This is often a serious issue, as readers prefer to mark the pages from where they must resume their study the next time.

There are some problems in this scenario –

1. User can have the books issued in his name (for a fixed period of time)
2. Any delays result in penalties.
3. A book meant for a previous reader ends up with a new reader.
4. No marking of books so that a new reader can ensure if he will get interrupted due to the choice of book.
5. Poor statistical analysis availability that could help the administration in scaling the library infrastructure as the demand grows.

There is a need for a unified management system for the library that can synchronize with the administration and the subscriber data in real time.

Scope

The library management system will be an online interface that will let the user as well as the administration, handle the library activities effectively.

1. A user can mark the book (by book ID); ensuring that the new reader will return the book to the bookshelf before the previous reader arrives.
2. No more penalties, as the books will stay in the library and available for the reader whenever he wishes to read them.
3. Administration will have the complete data to analyze if scaling is required or the number of copies of a particular book are more in demand.

Overview

The Library Management System will tackle various issues which are well known and so common that they are no longer considered issues but a new normal.

The system will offer some benefits and will offer specific functions based on the user's requirements.

Some of the main functions offered by the system –

1. Login Panel
 - For Admin
 - For Subscribers
2. Addition of new subscriber
3. Updating in subscriber data
4. Subscriber activity monitoring
5. Feedback service
6. Data analysis (Most read books, least read book, total subscribers, active daily subscribers)
7. Addition of new books